



given

$$F = 400 \text{ lb}$$

$$E = 10.0 \times 10^6 \text{ psi}$$

$$\delta = 0.009 \text{ in}$$

$$d = 0.50$$

$$A = \frac{\pi d^2}{4} = \frac{\pi (0.5)^2}{4} = 0.19635 \text{ in}^2$$

$$\delta = \frac{FL}{EA} \quad \begin{array}{c} \text{to find} \\ \text{a Length} \end{array} \quad L = \frac{\delta EA}{F} = \frac{(0.009)(0.19635)(10.0 \times 10^6)}{400}$$

$$L = 44.18$$

Von Mises stress

$$\sigma_{\max} = 1.487 \times 10^7 \text{ Pa}$$

$$1 \text{ psi} = 6894.76 \text{ Pa}$$

$$\frac{1.487 \times 10^7 \text{ Pa}}{6894.76 \text{ Pa}} \left| \frac{1 \text{ psi}}{1} \right| = 2156.7 \text{ psi}$$

$$\frac{2156.7 \text{ psi}}{1000 \text{ psi}} \left| \frac{1 \text{ ksi}}{1} \right| = 2.157 \text{ ksi}$$

$$2.157 \text{ ksi} < 40 \text{ ksi}$$

well below yield strength

$$FS = \frac{S_y}{\sigma_{\max}} = \frac{40}{2.157} = 18.54$$

$$\delta_{\text{calculated}} = 0.009 \text{ in}$$

$$\delta_{\text{FEA}} = 0.2286 \text{ mm}$$

$$\frac{0.2286 \text{ mm}}{25.4 \text{ mm}} \times 1 \text{ in} = 0.009 \text{ in}$$

$$\% \text{ difference} = \frac{|\delta_{\text{FEA}} - \delta_{\text{calc}}|}{\delta_{\text{calc}}} \times 100 = 0\%$$

Pin-hole stress concentration

$$K_t = 3$$

$$\sigma = 2.157 \text{ ksi}$$

$$K_t = \frac{\sigma_{\text{max}}}{\sigma_{\text{nom}}} \Rightarrow \sigma_{\text{max}} = K_t \cdot \sigma_{\text{nom}}$$

$$\sigma_{\text{max}} = (3)(2.157) = 6.471 \text{ ksi}$$

$$FS = \frac{S_y}{\sigma_{\text{max}}} = \frac{40}{6.471} = 6.18$$